INCH-POUND
MS27709J
05 January 2022
SUPERSEDING
MS27709H
1 November 2011

MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, 10 AMPERES, 4 PDT, TYPE I, HERMETICALLY SEALED

Reactivated after 05 January 2022 and may be used for new and existing design acquisitions

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relays described herein shall consist of this specification sheet and MIL-PRF-6106.

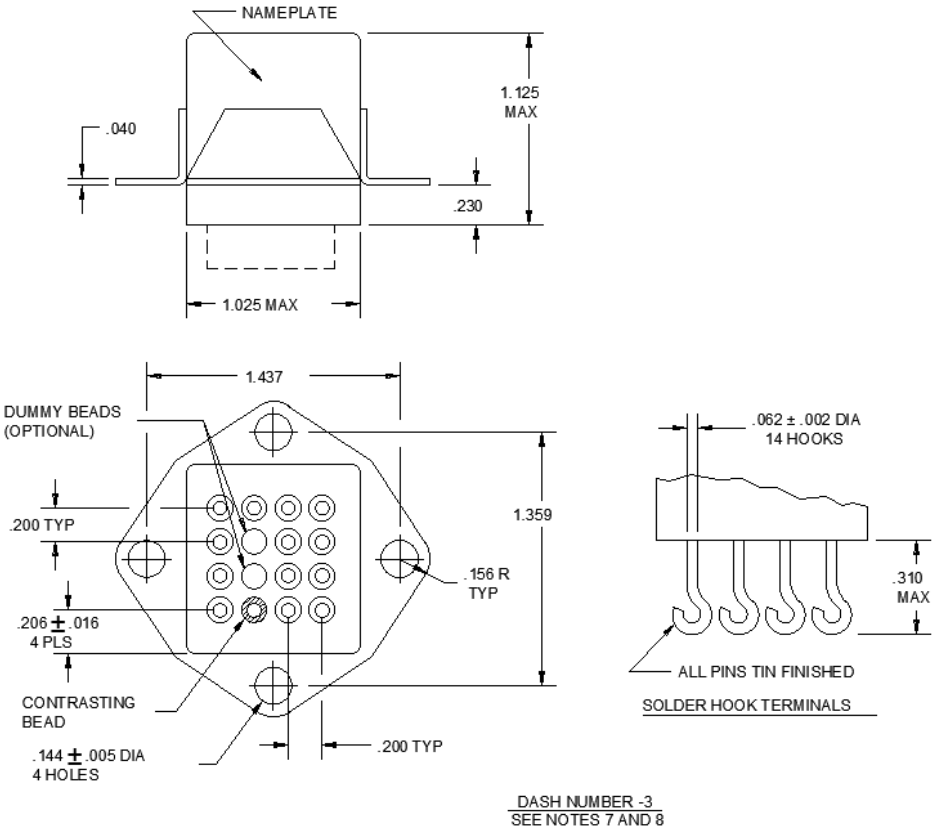


FIGURE 1. Configurations and dimensions.

AMSC N/A

FSC 5945

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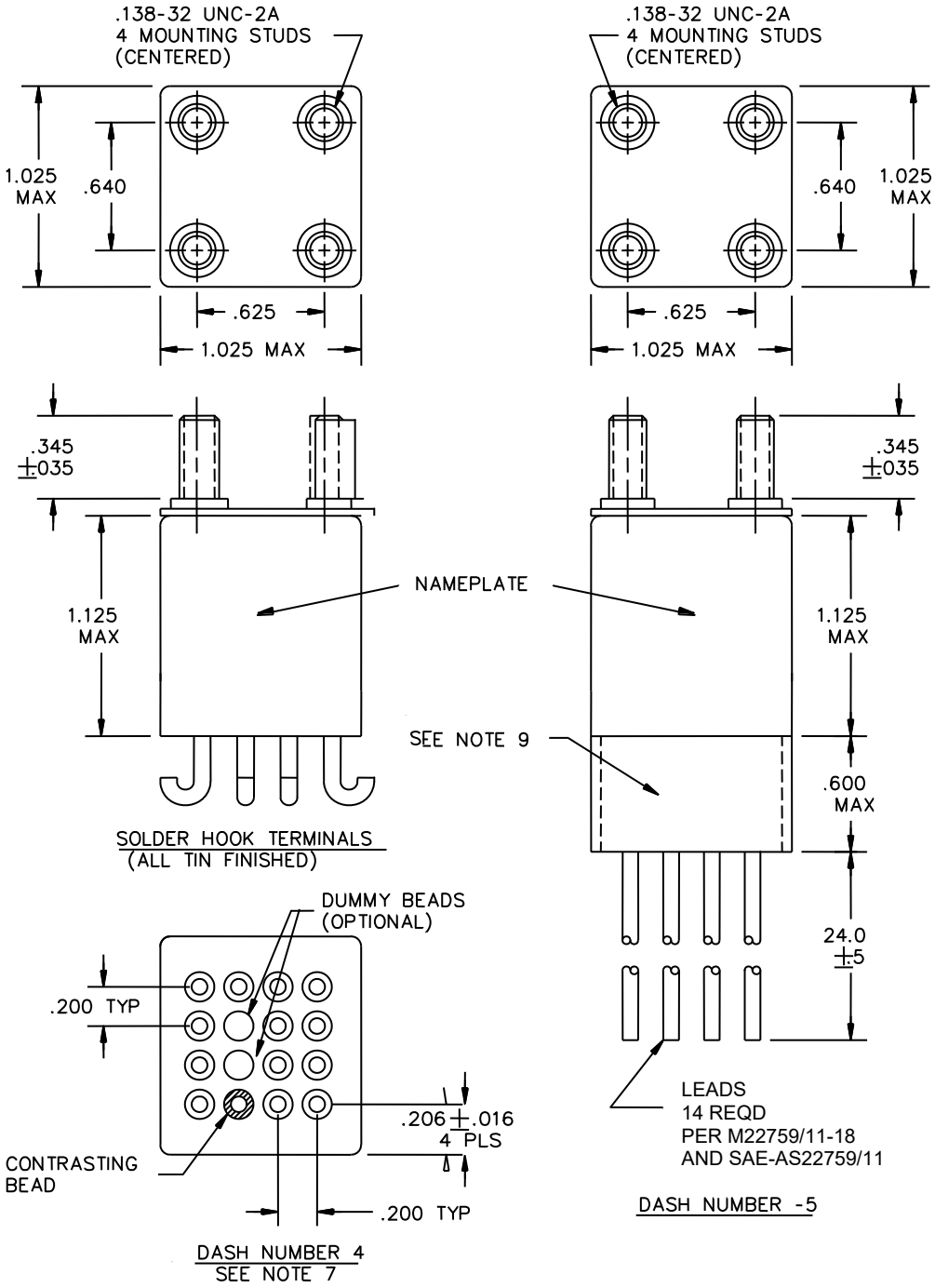


FIGURE 1. Configurations and dimensions - Continued.

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Inches	mm	Inches	mm	Inches	mm
.002	0.05	.156	3.96	.625	15.88
.005	0.13	.200	5.08	.640	16.26
.010	0.25	.206	5.23	1.025	26.04
.016	0.41	.230	5.84	1.125	28.58
.035	0.89	.300	7.62	1.359	34.52
.040	1.02	.310	7.87	1.437	36.50
.062	1.57	.345	8.76	24.0	610
.138	3.51	.500	12.70		
.144	3.66	.600	15.24		

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal numbers shall not appear on relay header. There shall be affixed to the relay legible circuit diagram that identifies each terminal location specified.
5. In the event of a conflict between the test of this document and the references cited herein, the text of this document shall take preference.
6. Referenced Government documents of the issue listed in that issue of the Acquisition Streamlining and Standardization Information System (ASSIST) specified in the solicitation forms a part of this specification sheet to the extent specified herein.
7. Orientation of solder lugs is optional.
8. Shape of flange is optional within the envelope dimensions shown.
9. Shell material shall be diallyl phthalate type SDG in accordance with ASTM-D5948, or equivalent material as approved by the qualifying activity. Shell shall be filled with encapsulation in accordance with MIL-PRF-23586, grade B, type II, class II, or equivalent filler as approved by the qualifying activity.

FIGURE 1. Configurations and dimensions - Continued.

Lead coding (-5 only)			
Term	Color	Term	Color
A1	Blue	C2	White-brown
A2	Brown	C3	White-red
A3	Red	D1	White-yellow
B1	Yellow	D2	White-orange
B2	Orange	D3	White-green
B3	Green	X1	White
C1	White-blue	X2	Black

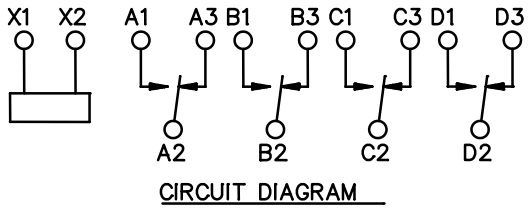


FIGURE 2. Lead coding and circuit diagram.

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REQUIREMENTS:

Configurations and dimensions: See figure 1.

Lead coding and circuit diagram: See figure 2.

Dash numbers and general characteristics: See table I.

Contact data:

Rated contact load: See table II.

Maximum contact drop, initial: 0.150 V.

After life test: 0.175 V.

Overload current (NO): 40 amperes dc.

Rupture current (NO): 50 amperes dc.

Operating characteristics: See table III.

Duty rating: Continuous.

RFI specification: Not applicable.

Electrical data:

Minimum insulation resistance:

Initial: 100 megohms.

After life or environmental test: 50 megohms.

Dielectric strength:

Sea level, 2-5 seconds:

	Initial	After life tests
Coil to case:	1,000 V rms	1,000 V rms
Auxiliary contacts:	N/A	N/A
All other points:	1,250 V rms	1,000 V rms

Altitude, 1 minute (80,000 ft.): 1/

Coil to case:	N/A	250 V rms
Auxiliary contacts:	N/A	N/A
All other points:	N/A	350 V rms

1/ Dielectric may be improved by suitable insulation of terminals and wiring after installation.

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Environmental characteristics:

Temperature range: -70°C to +125°C.

Maximum altitude rating: 80,000 feet.

Shock, g-level: 50 g's.

Duration: 10 ms.

Maximum duration contact opening: 1 ms.

Vibration, sinusoidal:

G-level: 20 g's.

Frequency range curve: 10 Hz to 2,000 Hz.

Vibration, random: Not applicable.

High shock: Not applicable.

Acceleration: 15 g's.

Terminal strength (high temperature pull and torque test): Not applicable.

Part or Identifying Number (PIN): MS27709- (plus applicable dash number from table I).

Quality assurance provisions: Group B and group C testing are not required. The manufacturer shall notify the qualifying activity in the event of any design or construction changes, and shall impose additional testing requirements as necessary.

Qualification by similarity: See MIL-PRF-6106.

Supersession data: See table IV.

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TABLE I. Dash numbers and general characteristics.

PIN MS27709-	Type	Coil	Terminal type	Mounting or mating socket	Max weight (pound)
3	I	dc	Solder hook	Flange (4 hole)	0.2
4	I	dc	Solder hook	Stud	0.2
5	I	dc	Potted leads	Stud	0.5

TABLE II. Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/				See appro-priate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	100	10	10			10				10				
Inductive	50	8	8			8				8				
Motor	100	4	4			4				4				
Lamp	100	2	2			2				2				
Transfer load														2/
Mechanical' life (reduced current	400	2.5	2.5			2.5				2.5				
Mixed loads		Applicable in accordance with MIL-PRF-6106.												

1/ Absence of value indicates that the relay is not rated for 3 phase applications.
2/ Transfer load indicates that the relay is suitable for transfer between unsynchronized ac power supplies at the rating indicated.

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TABLE III. Operating characteristics.

PIN MS27709-	Coil data											Time - (milliseconds maximum)					
	Coil	Rated			Max		Max pickup voltage			Hold voltage 2/	Drop-out voltage 2/	Oper-ate 3/	Re-lease 4/	Contact bounce			
														Main		Aux	
		Volts 1/	Freq. (Hz)	Res (Ω) min	Volts	A	Normal 2/	High temp test	Cont current test					NO	NC	NO	NC
3	X1,X2	28	dc	260	29	.200	18	19.8	22.5	7.0	1.5	15	15	2	2		
4	X1,X2	28	dc	260	29	.200	18	19.8	22.5	7.0	1.5	15	15	2	2		
5	X1,X2	28	dc	260	29	.200	18	19.8	22.5	7.0	1.5	15	15	2	2		

1/ CAUTION: Use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With rated coil voltage.
4/ From rated coil voltage.

TABLE IV. Supersession data.

Superseded PIN MS27709-	Replacement PIN M83536/15-
1	021
2	022
6	No replacement

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The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents. In addition to MIL-PRF-6106, this document references the following:
MIL-PRF-23586 ASTM-D5948 SAE-AS22759/11

Custodians:
Navy - AS
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5945-2022-005)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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